

# 17 Structures and Unions - Zero to Hero questions list

July 18, 2026

## 1 Structures and Unions in C — Practice Questions (50 Questions with Hints Only)

The questions are arranged from **Beginner** → **Intermediate** → **Advanced**.

---

### 2 Beginner Level (1–15)

#### 2.0.1 1. Create a structure to store student details.

**Hint:** Include name, roll number, and marks.

---

#### 2.0.2 2. Create a structure to store employee information.

**Hint:** Store ID, name, salary, and department.

---

#### 2.0.3 3. Create a structure to store book details.

**Hint:** Store title, author, price, and ISBN.

---

#### 2.0.4 4. Create a structure to store a person's information.

**Hint:** Store name, age, gender, and city.

---

#### 2.0.5 5. Create a structure to store product details.

**Hint:** Store product ID, name, price, and quantity.

---

**2.0.6 6. Read and display the details of one student.**

**Hint:** Use structure variables and `scanf()/printf()`.

---

**2.0.7 7. Read and display details of five students.**

**Hint:** Use an array of structures.

---

**2.0.8 8. Find the average marks of a student.**

**Hint:** Store marks of three subjects in the structure.

---

**2.0.9 9. Find the student with the highest marks.**

**Hint:** Compare total marks of all students.

---

**2.0.10 10. Store and display details of multiple employees.**

**Hint:** Use an array of structures.

---

**2.0.11 11. Create a structure to store date (DD/MM/YYYY).**

**Hint:** Separate day, month, and year fields.

---

**2.0.12 12. Create a structure to store time (HH:MM:SS).**

**Hint:** Use integer members.

---

**2.0.13 13. Create a structure to represent a point in 2D space.**

**Hint:** Store x and y coordinates.

---

**2.0.14 14. Create a structure to represent a rectangle.**

**Hint:** Store length and breadth.

---

**2.0.15 15. Calculate the area of a rectangle using a structure.**

**Hint:** Pass the structure to a function.

---

### **3 Intermediate Level (16–35)**

**3.0.1 16. Pass a structure to a function and display its contents.**

**Hint:** Pass by value.

---

**3.0.2 17. Pass a structure to a function and modify its values.**

**Hint:** Pass by pointer.

---

**3.0.3 18. Return a structure from a function.**

**Hint:** Return the entire structure.

---

**3.0.4 19. Compare two structures manually.**

**Hint:** Compare each member individually.

---

**3.0.5 20. Create an array of structures and sort students by marks.**

**Hint:** Use Bubble Sort.

---

**3.0.6 21. Sort employees by salary.**

**Hint:** Swap complete structures.

---

**3.0.7 22. Search a student by roll number.**

**Hint:** Perform a linear search.

---

**3.0.8 23. Search an employee by employee ID.**

**Hint:** Compare IDs sequentially.

---

**3.0.9 24. Count students scoring above 75%.**

**Hint:** Calculate percentage first.

---

**3.0.10 25. Store subject marks inside a nested structure.**

**Hint:** Use one structure inside another.

---

**3.0.11 26. Create a nested structure for an address.**

**Hint:** House number, street, city, state, PIN.

---

**3.0.12 27. Store student details along with address.**

**Hint:** Use nested structures.

---

**3.0.13 28. Create a structure containing an array.**

**Hint:** Store multiple marks inside one structure.

---

**3.0.14 29. Create a structure containing another structure and display all details.**

**Hint:** Access nested members using the dot operator.

---

**3.0.15 30. Find the youngest employee.**

**Hint:** Compare ages.

---

**3.0.16 31. Find the oldest person in a list.**

**Hint:** Traverse the array once.

---

**3.0.17 32. Calculate total salary of all employees.**

**Hint:** Sum the salary field.

---

**3.0.18 33. Calculate the average salary.**

**Hint:** Divide total salary by employee count.

---

**3.0.19 34. Store and display details of books using structures.**

**Hint:** Use an array of book structures.

---

**3.0.20 35. Find the most expensive book.**

**Hint:** Compare the price field.

---

## **4 Structures with Pointers (36–42)**

**4.0.1 36. Access structure members using a pointer.**

**Hint:** Use the `->` operator.

---

**4.0.2 37. Dynamically allocate memory for a structure.**

**Hint:** Use `malloc()`.

---

**4.0.3 38. Dynamically allocate memory for an array of structures.**

**Hint:** Allocate memory for multiple records.

---

**4.0.4 39. Pass a structure pointer to a function.**

**Hint:** Modify the original data.

---

**4.0.5 40. Create a linked list node using a structure.**

**Hint:** Include a pointer to the next node.

---

**4.0.6 41. Create a self-referential structure.**

**Hint:** A structure containing a pointer to its own type.

---

**4.0.7 42. Store student records using dynamically allocated structures.**

**Hint:** Combine `malloc()` with arrays or linked lists.

---

## 5 Unions (43–47)

**5.0.1 43. Create a union containing an integer, float, and character.**

**Hint:** Observe shared memory usage.

---

**5.0.2 44. Demonstrate how a union stores only one value at a time.**

**Hint:** Assign different members one after another.

---

**5.0.3 45. Compare the memory size of a structure and a union.**

**Hint:** Use the `sizeof` operator.

---

**5.0.4 46. Store different types of data using a union.**

**Hint:** Access only the last assigned member.

---

**5.0.5 47. Create a structure containing a union.**

**Hint:** Use the union for flexible data storage.

---

## 6 Advanced Level (48–50)

**6.0.1 48. Create a Student Management System using structures.**

**Hint:** Implement:

- Add Student
  - Display Students
  - Search Student
  - Update Student
  - Delete Student
-

**6.0.2 49. Create an Employee Payroll System using structures.**

**Hint:** Calculate:

- Basic Salary
  - HRA
  - DA
  - Gross Salary
  - Net Salary
- 

**6.0.3 50. Build a Library Management System using structures and unions.**

**Hint:** Divide the program into functions such as:

- Add Book
- Display Books
- Search Book
- Issue Book
- Return Book
- Delete Book
- Update Book Details
- Exit